

Fourier analysis – 1

Fourier series: convergence and summation

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Theoretical minimum.

Definition. For $f \in L^1[-\pi, \pi]$ put $c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt$ and $S_N f(x) = \sum_{|n| \leq N} c_n e^{inx}$. Then $S_N f = f * D_N$, where $D_N(t) = \sum_{|n| \leq N} e^{int} = \frac{\sin((N+\frac{1}{2})t)}{\sin \frac{t}{2}}$ is the Dirichlet kernel: everything about convergence is a statement about this convolution.

Theorem 1 (localisation and Dini). Whether $S_N f(x_0)$ converges, and to what, depends only on the values of f near x_0 : if f vanishes on an interval, its Fourier series converges to zero there. Dini's criterion: if $\int_0^\pi \frac{|f(x_0+t)+f(x_0-t)-2s|}{t} dt < \infty$, then $S_N f(x_0) \rightarrow s$; in particular this holds at every point of Hölder continuity. Jordan's criterion: for f of bounded variation the series converges to $\frac{f(x_0+)+f(x_0-)}{2}$.

Theorem 2 (summation by positive kernels). The Fejér means $\sigma_N f = f * F_N$ and the Abel–Poisson means $f * P_r$ converge to f uniformly for continuous f , because F_N and P_r are approximate identities: non-negative, of total mass one, concentrating at zero. The Dirichlet kernel is not non-negative and $\|D_N\|_{L^1} \sim \frac{4}{\pi^2} \ln N$ — the Lebesgue constants computed in Trig-6 — and this single fact is the source of all the pathologies of pointwise convergence.

Fact 1 (divergence). Since the Lebesgue constants are unbounded, the Banach–Steinhaus principle (Functional sequences, series and spaces – 4) produces a continuous function whose Fourier series diverges at a prescribed point. Kolmogorov built an L^1 function whose series diverges everywhere; Carleson proved that for $f \in L^2$ the series converges almost everywhere.

Fact 2 (Gibbs). Near a jump the partial sums overshoot the jump by the

factor $\frac{2}{\pi} \int_0^\pi \frac{\sin t}{t} dt - 1 \approx 0.0895$, independently of the function: the overshoot does not disappear as $N \rightarrow \infty$, it only moves closer to the jump.

1. Prove that $S_N f = f * D_N$ with the closed form of D_N , prove the localisation principle and Dini's criterion, and deduce that the Fourier series of a function Hölder continuous at a point converges at that point to its value.
2. Prove Fejér's theorem for a continuous 2π -periodic function and its Abel–Poisson analogue, and deduce the uniqueness theorem: two continuous functions with the same Fourier coefficients coincide.
3. Prove that there exists a continuous function whose Fourier series diverges at the point 0. (Use the growth of the Lebesgue constants from Trig-6 and the Banach–Steinhaus principle from Functional sequences, series and spaces – 4.)
4. (Kapacëb 1.68) Let a continuous function be represented by a convergent Fourier series $f(x) = \sum_{n \geq N} (a_n \cos nx + b_n \sin nx)$ with $N \geq 1$. Prove that f has at least $2N$ zeros on the period $[0, 2\pi)$.
5. (S.T. Yau CSMC 2016) Let f be Riemann integrable on $[-\pi, \pi]$ with Fourier series $\sum_{n \in \mathbb{Z}} a_n e^{inx}$ and suppose $|a_n| \leq \frac{K}{|n|}$ for all $n \neq 0$. Show that

$$\left| \sum_{n=-N}^N a_n e^{inx} \right| \leq \sup_{y \in [-\pi, \pi]} |f(y)| + 2K$$

for all x and all N .